

AQA A-Level Chemistry Paper 2: Student Strategy Guide

Based on 10 Past Papers + 8 Examiner Reports (Specimen 2015, 2017-2025)

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Important disclaimer This guide is based on HSJ Tuition's independent analysis of all publicly available AQA Chemistry Paper 2 (7405/2) question papers and examiner reports from 2015 to 2025. We are highly confident in the patterns and advice presented here: they are grounded in ten papers' worth of examiner evidence, not guesswork. However, **exam content cannot be predicted or guaranteed**. AQA can set any topic from the specification at any time. This guide is a revision strategy tool designed to help you prioritise your time and eliminate common errors. It is not a definitive prediction of what will appear. Use it alongside full specification coverage, not instead of it. HSJ Tuition accepts no responsibility for topics not appearing or appearing differently than anticipated.

WHY THIS GUIDE EXISTS

This guide was built by analysing every AQA Chemistry Paper 2 (7405/2) from the specimen paper through June 2025, plus every examiner report published for this paper. It tells you **exactly where students lose marks**, not based on opinion, but on what examiners have flagged year after year.

WHAT PAPER 2 ASSESSES

Paper 2 is a 2-hour written exam worth 105 marks (35% of the A-level). It consists of short and long answer questions. Typically 9-11 questions, always 1 Level of Response question (6-7 marks).

According to the AQA specification, Paper 2 assesses:

- **Relevant physical chemistry** (sections 3.1.2 to 3.1.6 and 3.1.9)
- **Organic chemistry** (section 3.3)
- **Relevant practical skills**

This means the physical chemistry on Paper 2 covers: Amount of substance, Bonding, Energetics (AS), Kinetics (AS and A2), and Chemical equilibria (Kc). Topics like Born-Haber cycles, electrode potentials, buffers, K_a , K_p , entropy, and Gibbs free energy sit in different specification sections and are assessed on Paper 1 only.

Paper 2 = **rates + Arrhenius + Kc + organic chemistry + spectroscopy + practical skills**.

PART 1: WHAT TO REVISE FIRST

These topics appear **every year** and together make up roughly 60-75 marks:

Priority	Topic	Frequency	Marks per paper
1	Rate equations: orders, k, units	10/10	6-10
2	¹ H NMR: shifts, splitting, integration, structure	9/10	5-8
3	IR spectroscopy: functional group ID	9/10	2-4
4	Arrhenius equation: E_a from graph/calculation	8/10	3-5
5	Kc expression and calculation	8/10	5-8

These topics also have **high frequency** (nearly every year):

Priority	Topic	Frequency
6	¹³ C NMR: peak counting, symmetry	8/10
7	Nitration mechanism (curly arrows)	7/10
8	Aliphatic amine chemistry	7/10
9	Percentage yield calculation	7/10
10	Amino acids, zwitterions, peptide bonds	6/10

Do not skip the Tier 1 topics for the Tier 2 topics. A student who nails rates, NMR, IR, Arrhenius, and Kc recovers more marks than one who broadly covers everything.

PART 2: THE GOLDEN RULES

These rules are based on errors that appear in **multiple examiner reports**. They apply across the paper.

Rule 1: Derive k units algebraically every time

Do not memorise units. Rearrange $\text{rate} = k[A]^m[B]^n$, substitute units, cancel. Wrong unit powers are one of the most common errors across all papers (flagged 2017, 2018).

Rule 2: Curly arrows start from lone pairs or bonds, never from atoms

An arrow starting from an atom (rather than its lone pair or a bond) scores zero for that step. Flagged 2017, 2018, 2022, 2024.

Rule 3: "Observations" are what you SEE

"A precipitate forms" = observation. "AgCl is produced" = deduction. When asked to "state your observation", give colours, precipitates, effervescence. Never chemical names.

Rule 4: KCN is the reagent, not CN-

In nucleophilic addition of HCN, the reagent is KCN (or NaCN) in dilute acid. Writing "CN-" alone is penalised as incomplete (flagged 2017, 2018).

Rule 5: Explain splitting patterns properly

Do NOT just say "n+1 rule". You must state: "The H atoms on this carbon are adjacent to n H atoms on the neighbouring carbon, so the signal is split into n+1 peaks." Explicitly flagged 2017 (Q10.5).

Rule 6: Account for symmetry in ^{13}C NMR

Equivalent carbons give ONE peak. Students routinely predict double the correct number. Flagged 2017 (Q10.4).

Rule 7: Racemic mixture needs TWO facts

(1) The carbonyl carbon is planar (sp^2). (2) The nucleophile attacks from either face with equal probability. Both required. "Two products form" alone does not score (flagged 2017 Q04.3).

Rule 8: "Refracted" scores zero for optical isomers

Plane-polarised light is **rotated**, not "refracted" or "bent". Called out 2017 (Q04.2).

Rule 9: Significant figures match the data

If data is to 3 sig figs, your answer must be too. Penalised 2017 (Q05.4), 2018 (Q04.3).

Rule 10: Negate the Arrhenius gradient

Gradient of $\ln k$ vs $1/T$ = $-E_a/R$. So E_a = **-gradient** x R. Students forget to negate and get a negative E_a . Flagged multiple years.

PART 3: TOPIC-BY-TOPIC KEY MISTAKES**Rates and Kinetics**

- **Units of k**: wrong power signs ($\text{dm}^{-3} \text{mol}$ instead of $\text{mol}^{-1} \text{dm}^3$). Derive algebraically every time.
- **Tangent drawing**: tangent drawn too small gives imprecise gradient. Extend it across as much of the graph as possible (flagged 2017 Q02.1).
- **Arrhenius**: reading $\ln A$ as A (it is the natural log, not A itself). Forgetting to convert kJ to J. Misreading the $1/T$ axis. All flagged 2017 Q03.2.
- **Rate-determining step**: must link the species in the slow step to the species in the rate equation. Just saying "it is the slowest step" is not enough.
- **Half-life from graph**: reading at the wrong concentration level, or confusing the order from the half-life pattern.
- **Zero order from graph**: zero order gives a straight-line graph with negative gradient. The rate is constant, independent of concentration. Tested 2024 (Q1).

Kc Equilibrium

- **Including water or solids** in the K_c expression: loses marks. Only dissolved species in square brackets (flagged NOV20 Q09.3).
- **ICE table**: using wrong stoichiometric ratios in the "Change" row. Forgetting to divide moles by volume to get concentrations (flagged 2018 Q04.3).
- **Dilution effect**: adding water dilutes all concentrations. Equilibrium shifts to the side with more moles of solute. K_c does NOT change. 65.8% of students failed this in 2018 (Q04.4).
- **Sig figs**: giving K_c to wrong number of sig figs (flagged 2017, 2018).
- **Only temperature changes K_c** : catalyst has no effect on K_c . Tested almost every year.

Nitration Mechanism

- **Missing the horseshoe arrow** showing the pi system attacking NO₂⁺. Do NOT show a lone pair on NO₂⁺: it is electron-deficient (flagged 2017 Q08.2).
- **Drawing the arenium ion** without the + charge, or without showing H still attached to the carbon bearing NO₂⁺ (flagged 2024 Q10.3).
- **Starting from nitrobenzene** when the question asks about benzene (flagged 2017).
- **"Hydration"** instead of **"hydrogenation"** of benzene: scores zero (flagged 2018 Q06.1).

Friedel-Crafts

- **Missing charges**: must show AlCl₄⁻ with its negative charge, and the + charge on the acylium ion RCO⁺ (flagged 2018 Q07.1).

Carbonyl Chemistry

- **H₂CN/NaBH₄**: racemic mixture explanation must specify the **carbonyl carbon** is planar, not "the molecule" (flagged 2017 Q04.3).
- **2,4-DNPH identifies C=O but does NOT distinguish aldehyde from ketone**. Use Tollens' or Fehling's for that.
- **Fehling's/Tollens'**: must describe the observation (silver mirror, brick-red precipitate), not name the product.

Acyl Chlorides

- HCl produced gives **misty/steamy fumes**, not a "white precipitate" (flagged 2018 Q07.3).
- Mechanism is **nucleophilic addition-elimination**, not "nucleophilic substitution" (different name).

Amine Basicity

- **Phenylamine**: must say "the lone pair on nitrogen is delocalised into the pi system of the benzene ring". Saying vaguely "the lone pair is delocalised" is not enough (flagged 2017 Q11.1).
- **Inductive effect**: alkyl groups donate electron density by positive inductive effect. Say "inductive effect", not just "electron donating".
- Aromatic amines are used to **manufacture** dyes, not used **as** dyes directly (flagged 2024 Q10.5).

Amino Acids and Peptides

- **Zwitterion**: do NOT remove H from the alcohol side chain (e.g. serine's -CH₂OH). Only the COOH group loses H to form COO⁻ (flagged 2017 Q06.2).
- **Two dipeptides**: A-B and B-A. If the question says "draw both", draw both (flagged 2017 Q06.3).
- **Extra oxygen in amide link**: drawing -COOH-NH- instead of -CO-NH- (flagged 2017).
- **Disulfide bridges** are covalent bonds between cysteine side chains. They are the strongest interaction in tertiary structure. Very few students know about them (flagged 2018 Q08.2).

NMR

- **Splitting explanation**: "n+1 rule" alone does not score. Must explain what n refers to for the specific signal (flagged 2017 Q10.5, 2018 Q11.1).
- **¹³C symmetry**: symmetric molecules have fewer peaks. Students predict double (flagged 2017 Q10.4).
- **¹H NMR for symmetric CH₂ groups**: two equivalent CH₂ groups give a singlet, not a triplet (flagged 2017 Q10.5).
- **Combined spectra deduction**: identify fragments individually, then assemble into ONE consistent structure. Students often get fragments right but cannot combine them (flagged 2020 Q07, 2022 Q06).

DNA

- **Nucleotide drawing**: must include the phosphate group and the CH₂ link between sugar and phosphate. Do not show two phosphate groups (flagged 2018 Q09.3).
- **Base pairs**: A-T has 2 hydrogen bonds, G-C has 3. Do not add extra H-bonds (flagged 2017 Q09.2).

Practical Skills

- **Melting point**: impure sample has a **lower AND broader** melting range. Both facts required (flagged 2020 Q04.6).
- **Recrystallisation**: cold water wash removes **soluble** impurities. Hot filtration removes **insoluble** impurities.
- **Yield calculation**: check the limiting reagent first when two reagents are given (flagged 2020 Q04.5).
- **Calorimetry**: m in q = mcΔT is mass of **WATER**, not mass of the fuel (flagged 2018 Q03.4). Negative sign for exothermic (flagged 2024 Q11.1).

PART 4: THE EASY MARKS CHECKLIST

Before you hand in your paper, check these **every time**:

- ☐ k units derived **algebraically** (not from memory)
- ☐ All curly arrows start from **lone pairs or bonds**
- ☐ Arrhenius: gradient **negated** before calculating E_a
- ☐ K_c expression: no water, no solids, correct powers
- ☐ NMR splitting: explained with **number of adjacent H**, not just "n+1"
- ☐ ¹³C NMR: checked **symmetry** to avoid double-counting peaks
- ☐ Racemic mixture: stated **planar carbonyl AND equal attack from both faces**
- ☐ All observations are **what you see** (colours, precipitates, gas), not deductions
- ☐ Significant figures match the data in the question
- ☐ Crossed out any work you do not want marked

These 10 checks alone are worth an estimated 8-12 marks per paper.

PART 5: TRENDS AND TOPIC PATTERNS

What 2025 Paper 2 covered (less likely in identical form)

- Free radical substitution mechanism (Q1)
- Electrophilic addition, Markovnikov (Q2)
- Nylon 6,6 + nylon 6 mechanisms (Q3)
- Recrystallisation procedure (Q4)
- Nitration mechanism - TNT context (Q5)
- NMR of amino acid in D₂O, protein structure (Q6)
- NMR structure determination - Level of Response (Q7)
- Ideal gas equation, percentage uncertainty (Q8)
- Arrhenius graph plotting (Q9)

Topics absent for multiple years

Topic	Last appeared	Gap
DNA / nucleotide	JUN19	6 years
TLC / chromatography	JUN19	6 years
Amine basicity as LoR	JUN17	8 years
Benzene stability as LoR	JUN18	7 years
K _c algebraic ICE (solve for x)	NOV21	4 years
Alkaline ester hydrolysis mechanism	JUN22	3 years
Protein as a large standalone Q	JUN22	3 years
HCN nucleophilic addition standalone	JUN22	3 years
Diazotisation / azo coupling	Never	In spec

Topics that appear every year (never skip)

- Rate equations: orders, k, units (10/10)
- ¹H NMR (9/10)
- IR spectroscopy (9/10)
- Some mechanism with curly arrows (9/10)
- ¹³C NMR (8/10)
- Arrhenius equation (8/10)
- K_c expression / calculation (8/10)

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